

Anurag Yadav

Hatiali Siding, Chabua, Dist.-Dibrugarh, Assam-786184 | +91-6000147471 | abcanuragyadav251@gmail.com

linkedin.com/Anurag-Yadav | github.com/Anurag-Yadav

Professional Summary

Results-driven M.Tech graduate in Artificial Intelligence and Machine Learning with research and industry experience in NLP, deep learning, computer vision, and predictive analytics. Published researcher with work presented at ICCNT 2025 and national conferences on abusive language detection, diabetic retinopathy severity assessment, and stock market forecasting. Skilled in Python, PyTorch, TensorFlow, SQL, and end-to-end AI pipeline development, from data acquisition and feature engineering to model deployment. Strong analytical and problem-solving abilities with a passion for advancing AI-driven research and innovation.

Education

DUIET, Dibrugarh University

Master of Technology in Computer Science and Engineering – AI & ML | CGPA: 8.25

2024 – Present
Dibrugarh, Assam

- Coursework: Mathematics for Machine Learning, Advanced Data Structures, Data Warehousing and Pattern Mining, Data Visualization and Machine Learning, Research Methodology and IPR, Optimization Technique, Deep Neural Network, Artificial Intelligence & Knowledge Representation, Natural Language Processing, Reinforcement Learning.

The Assam Kaziranga University

Bachelor of Technology in Computer Science and Engineering | CGPA: 8.08

April 2024
Jorhat, Assam

- Coursework: Database Management System, Unix and Linux Programming, Operating Systems, Artificial Intelligence, Fundamentals of Web Services in Cloud, Computer Organization and Architecture, Computer Networks, Compiler Design, Object Oriented Programming in Java, Machine Learning, Data Mining, Deep Learning, Design Analysis of Algorithms.

Kendriya Vidyalaya Chabua

Class 12 (Science) | Percentage: 81.6%

May 2020
Chabua, Dibrugarh, Assam

- Subjects: English Core, Physics, Chemistry, Mathematics, Computer Science, Physical Education.

Hemalata Handiqui Memorial Institute

Class 10 | Percentage: 90.8%

May 2018
Jorhat, Assam

- Subjects: English Comm., Hindi Course-B, Mathematics, Science, Social Science, Foundation of IT.

Skills

Programming: Python, C, C++, JavaScript, HTML, CSS, Prolog, MySQL

Frameworks / Libraries: PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, FastAPI, OpenCV, YOLO

Platforms & Tools: Git, Figma, Adobe Suite, Windows

Familiar With: Power BI, UI/UX Design, Android Development

Internship Experience

CSIR – North East Institute of Science and Technology (NEIST)

Machine Learning Intern

Jul 2025
Jorhat, Assam

- Worked on “Study of Land Use Land Cover (LULC) of Majuli District, Assam” under the guidance of Dr. V. M. Tiwari (Director) and Dr. Hridoy Jyoti Mahanta (Scientist, CSIR-NEIST).
- Built a Landsat-based dataset (2000–2025) to analyze land use and land cover change across Majuli Island using ArcGIS 10.8.

Cranes Varsity Pvt. Ltd.

Data Science and AI Intern

Aug–Sep 2023 (Remote), Feb–Mar 2024 (On-site)
Bengaluru, Karnataka

- Completed modules on RDBMS, Python fundamentals, advanced Python, data science, and artificial intelligence.
- Analyzed real-world datasets (infant mortality rate, obesity, etc.) and built a plagiarism detection tool as a minor project.

Acmegrade

Machine Learning Intern

Jul–Sep 2022
Remote

- Completed modules on data analysis and machine learning algorithms including Random Forest, KNN, SVM, and Regression.
- Built a cancer prediction system and a movie recommendation system using Random Forest and SVM.

Research Works

- Hybrid Attention-Based Sequence Models for Abusive Language Detection in Hindi
- A Dual-Model Approach for Diagnosing Diabetes Types and Assessing Diabetic Retinopathy Severity

- Deep Ensemble Learning for Stock Market Forecasting with Risk-Tolerant Investment Strategies and Uncertainty Quantification

Projects

GenScene: A Unified Neural-Symbolic Framework for Visual Scene Understanding

Python, PyTorch, Mask R-CNN, BLIP, CLIP, OpenCV, NetworkX, Neural-Symbolic AI

Developed GenScene, a unified Neural-Symbolic AI framework for visual scene understanding by integrating instance segmentation, image captioning, zero-shot relational reasoning, scene graph generation, and symbolic inference. Fine-tuned Mask R-CNN on the MS-COCO dataset for object detection and segmentation and utilized BLIP for multimodal caption generation. Implemented CLIP-based zero-shot relationship prediction to infer semantic interactions among detected objects and construct scene graphs. Designed a symbolic reasoning engine to derive high-level contextual knowledge from object relationships, enhancing explainability and interpretability. Extended the framework with English-to-Assamese multilingual scene description generation and evaluated its performance using mAP, mIoU, BLEU, ROUGE-1, and ROUGE-L metrics.

Study of Land Use Land Cover (LULC) of Majuli District, Assam

ArcGIS 10.8, Remote Sensing, GIS Analysis

Worked on a project titled “Study of Land Use Land Cover (LULC) of Majuli District, Assam”. The main aim of the project was to find the land use and land cover of Majuli Island in Assam using Landsat images from the years 2000–2025. A shape file for the Majuli location was created in Google Earth Pro and used as the outline figure. The dataset was built using Landsat imagery downloaded from USGS Earth Explorer and then rendered to determine the LULC of Majuli using ArcGIS 10.8.

Stock Market Price Prediction and Risk-Tolerant Investment Advisory System

Python, Deep Learning, LSTM, BiLSTM, TCN, NLP

Developed a deep learning-based stock market forecasting and investment advisory system using historical NASDAQ data, technical indicators (SMA, EMA, MACD, ATR, RSI, and OBV), financial news sentiment, and macroeconomic factors. Implemented and compared LSTM with Attention, BiLSTM with Multi-Head Attention, and Temporal Convolutional Network (TCN) models for stock price prediction and trend analysis. Evaluated predictive performance using MAE, RMSE, MSLE, R^2 , and Explained Variance Score, and generated personalized investment recommendations based on varying risk-tolerance profiles.

Dual-Model System for Diabetes Diagnosis and Diabetic Retinopathy Grading

Python, Machine Learning, Deep Learning, ResNet-18

Developed a dual-model AI framework for diabetes prediction and diabetic retinopathy (DR) severity classification by integrating machine learning and deep learning techniques. Applied Random Forest, CatBoost, Logistic Regression, KNN, and Naïve Bayes to clinical data containing glucose levels, BMI, insulin levels, and age for diabetes risk prediction. Implemented a pretrained ResNet-18 model to analyze retinal fundus images and classify diabetic retinopathy into five severity levels. Achieved 78.9% accuracy, demonstrating the effectiveness of AI-driven healthcare solutions for early disease diagnosis and clinical decision support.

Abusive Language Detection in Hindi

Python, NLP, Word2Vec, FastText, BiLSTM, Transformers

Developed a deep learning-based NLP system for detecting abusive language in Hindi social media content using a dataset of 41,831 labeled tweets. Applied text preprocessing techniques including normalization, tokenization, and lemmatization, and generated semantic representations using Word2Vec and FastText embeddings. Implemented BiLSTM with Multi-Head Attention and Transformer-Enhanced BiLSTM architectures for classification, achieving approximately 76% accuracy and demonstrating effective abusive content detection in a low-resource language setting.

Cancer Prediction System

Python, Scikit-learn, Random Forest, SVM, XGBoost

Designed and implemented a cancer prediction system that analyzes clinical diagnostic features to classify tumors. Applied data preprocessing, SMOTE-based class balancing, feature importance analysis, and machine learning algorithms including Random Forest, SVM, and XGBoost. Evaluated model performance using accuracy, confusion matrix, and cross-validation techniques.

Movie Recommendation System

Python, Machine Learning, Pandas, NumPy, Scikit-learn

Developed a machine learning-based movie recommendation engine using user rating patterns and movie metadata to provide personalized content suggestions. Implemented data preprocessing, feature extraction, and similarity-based recommendation techniques to identify user preferences and predict relevant movies. Evaluated recommendation performance through ranking and accuracy metrics, enhancing recommendation quality and user engagement.

Music Scale Recommender System

Python, Librosa, NumPy, Audio Signal Processing

Designed and implemented an intelligent music recommendation system that analyzes audio files to identify musical keys and suggest compatible scales. Utilized chroma feature extraction, signal preprocessing, and key-profile matching techniques to improve key detection accuracy and support music learning, composition, and improvisation.

AidSync

React Native, Figma

Made a React-based app named AidSync as my BTech final project. This app connects homeless people to NGOs to help them, with features like donations and complaint reporting.

PlagCheck

Python, Tkinter, DiffLib

Created a plagiarism checker tool in Python using Tkinter and DiffLib that examines and compares two text files by finding similarities between them.

Achievements & Publications

- **ICCCNT 2025** – Paper Presentation: Scalable AI system for real-time abusive content detection in Hindi using attention and transformer-based architectures, outperforming existing baselines on accuracy and F1 score.
- **National Conference on Multidisciplinary Approaches in Networks and Computing (NCMNC 2025)** – Presented research on AI and ethnobotany at a multidisciplinary forum.
- **National Conference on Integrating Ethnobotany and AI for Future Healthcare Solutions (Viksit Bharat@2047)** – Presented an integrated AI pipeline for early diabetes diagnosis and diabetic retinopathy severity assessment.

Courses & Certifications

Data Structures and Algorithms (IBM) • Python Bootcamp (Udemy) • Full Stack Web Development (Udemy) • Machine Learning Services with Watson Studio (IBM) • Database Management System (IBM) • SQL and Relational Databases 101 (IBM) • Predictive Modelling with IBM SPSS Modeler • Data Visualization with Python (IBM)

Languages

English (Professional) • Hindi (Professional) • Assamese (Professional) • Bhojpuri (Native)